FACULTY OF SCIENCE – UNIVERSITY OF PERADENIYA
DEPARTMENT OF STATISTICS AND COMPUTER SCIENCE

END SEMESTER EXAMINATION (SEMESTER I) – 2019/2020

ST103 – STATISTICS APPLICATIONS I (1 CREDIT)

Time allowed: **Two hours** Answer **ALL** questions

INSTRUCTIONS: You should prepare a text document (MS – WORD or Writer) including all the necessary outputs, answers and relevant interpretation. Save the file with the name **XXXXXX.doc** or **XXXXXX.odt** where **XXXXXX** represents your registration number (Eg: If the registration number is "**S/18/001**" then the file name should be "**S18001.doc**" or "**S18001.odt**")

Also save the R-studio project with the relevant R-scripts with the name "**XXXXXX.Rproj**" in your home directory. **XXXXXX** represents your registration number. Inside this project, create a separate R-script file for each question including all the R commands and save your R-script files with the file names below.

Question	1	2	3	4
R-script file	Que1.R	Que2.R	Que3.R	Que4.R

You are advised to save all your works regularly to avoid losses of data due to power failures.

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1. Consider the text file Crashes_UK in your home directory to answer the following questions. The data are from UK police-reported vehicle crashes and there are eleven variables as follows.

Variable	**Description**
Region	Region where the accident happened
Urban_or_Rural_area	Type of the region: Urban or Rural
Road_Class	Class of the road: A, B, C or Unclassified
Road_Surface_Condition	Surface condition of the road: Dry, Wet or Damp
Speed_Limit	Speed limits for the vehicles
Year	Year
Age_of_Driver	Experience of the driver
Age_of_Vehicle	Age of the vehicle
Vehicle_Make	Brand of the vehicle
Vehicle_Category	Type of the vehicle
Accident_Severity	Condition of the accident

   i. Import the Crashes_UK dataset in your home directory and rename it as ***Accidents***.

   ii. Using a suitable R command change the column names ***Urban_or_Rural_Area*** and ***Road_Surface_Condition*** to ***Region_Type*** and ***Road_Status***, respectively.

   iii. Obtain the summary statistics for ***Age_of_Driver*** and ***Age_of_Vehicle***, and interpret the results separately.

iv. Create a vector called ***ID*** containing numbers one to sixty with increment of one, and create a new data frame called ***New_accidents*** combining the ***ID*** vector and ***Accidents*** data frame.

v. Select the details of all car accidents happened under ***Dry*** road surface condition and store the results as ***Dry_accidents***

vi. Create a contingency table for the two categorical variables ***Road_class*** and ***Accident_Severity***.

vii. Draw two horizontal boxplots for ***Age_of_Vehicle*** according to the ***Region_Type***.

viii. Find the frequency of ***Vehicle_Make*** and create a bar plot using rainbow colours. Add a suitable title, x-axis name and y-axis name to the same plot.

2. a)

i. Create a 3×3 empty matrix $\boldsymbol{A}$ and a 2×3 empty matrix $\boldsymbol{B}$.

ii. Create the following vectors ***mvec1*** and ***mvec2***.

$$mvec1 = \{7, 2, 1, 0, 3, -1, -3, 4, -2\}$$
$$mvec2 = \{1, 0, 2, 1, 3, -2\}$$

Insert ***mvec1*** and ***mvec2*** as elements to the empty matrices $\boldsymbol{A}$ and $\boldsymbol{B}$, respectively.

iii. Do the matrices $\boldsymbol{A}$ and $\boldsymbol{B}$ have an inverse? Justify your answer.

iv. Add another row to matrix $\boldsymbol{B}$ with values $\{1, 2, 5\}$.

v. Find $\boldsymbol{A}'$, $\boldsymbol{B}'$, $\boldsymbol{A}+\boldsymbol{B}$, $\boldsymbol{A}-\boldsymbol{B}$ and $\boldsymbol{AB}$ (both element-wise multiplication and matrix multiplication)

vi. Find a matrix $\boldsymbol{C}$ such that $\boldsymbol{CA} = \boldsymbol{B}$. Round off the elements of $\boldsymbol{C}$ to a whole number.

vii. Create a 3×3 diagonal matrix $\boldsymbol{I}$ (identity matrix) with the value 1 in its diagonal.

viii. Prove the fact that $\boldsymbol{A}^{-1}\boldsymbol{A} = \boldsymbol{I} = \boldsymbol{A}\boldsymbol{A}^{-1}$.
Hint: Use *all.equal(X,Y)* function to check if *X* and *Y* are equal

b) Consider the following data vectors and answer questions given below.

$$x = \{4.0, 4.5, 5.0, 5.5, 6.0, 6.5, 7.0\} \text{ and } y = \{33, 42, 45, 51, 53, 61, 62\}$$

i. Find the matrix $\boldsymbol{X}$ such that

$$\boldsymbol{X} = \begin{bmatrix} n & \sum_{i=1}^{n} x_i \\ \sum_{i=1}^{n} x_i & \sum_{i=1}^{n} x_i^2 \end{bmatrix},$$

where n= number of elements in x.

ii. Find the matrix $\boldsymbol{Y}$ such that

$$\boldsymbol{Y} = \begin{bmatrix} \sum_{i=1}^{n} y_i \\ \sum_{i=1}^{n} x_i y_i \end{bmatrix}.$$

iii. Find the matrix $\boldsymbol{D}$ such that $\boldsymbol{D} = \boldsymbol{X}^{-1}\boldsymbol{Y}$.

iv. What is the dimension of $\boldsymbol{D}$?

v. Change the row names of ***D*** to ***beta_1*** and ***beta_2***. Change the column name of ***D*** to ***values***.

3. a) Create the below given sequence of vectors using the functions ***seq*** and ***rep*** when necessary.

i. $\{2, 4, 6, \dots, 16\}$

ii. $\{14, 12, 10, \dots, 2\}$

iii. $\{2, 4, 6, \dots, 14, 16, 14, 12, \dots, 2\}$

iv. $\{3, 3, \dots, 3, 9, 9, \dots, 9, 7, 7, \dots, 7\}$, where there are 10 occurrences of 3, 20 occurrences of 9 and 30 occurrences of 7.

v. $\{1, 2, 3, 1, 2, 3, \dots, 1, 2, 3\}$, where there are 10 occurrences of the set $\{1, 2, 3\}$

b) Create two data frames named ***Reg_Details*** and ***Marks*** and fill them with the following observations.

Reg_Details:

Index Number	**Name**	**Gender**
001	Jim	M
002	Mary	F
003	Arthur	M
004	Natali	F
005	Emma	F

Marks:

Index Number	**IT**	**Science**	**English**
001	56	61	78
002	86	82	84
003	64	67	60
004	90	93	82
005	45	34	57

i. Merge Reg_Details and Marks datasets and name the merged dataset as ***Merge_data***

ii. Change Jim's Science marks to 75 and Arthur's English marks to 81.

iii. Extract a subset named ***Female_stu*** from ***Merge_data*** such that it contains all the records of females (Gender=F)

iv. Plot a line graph for the three subject marks and give a suitable title, x-axis name and y-axis name.

v. Place a legend (indicating the subject) at the top-right corner with the frame using the same colours.

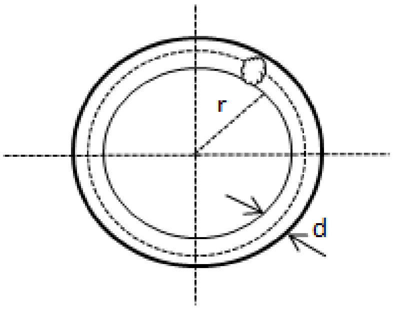
4. a) The surface area S of a ring in shape of a torus with an inner radius r and the diameter d is given by

$$S = \pi^2(2r + d)d$$

The ring is to be plated with a thin layer of coating. The weight W of the coating can be calculated approximately as

$$W = \gamma St,$$

where t is the thickness of the coating and γ is the specific weight of the coating material.



i. Write an R-function to calculates the weight of the coating by using the four input arguments r, d, t and γ.

ii. Using the R-function created in part (i), calculate the weight of a gold coating when $r = 0.9\ cm, d = 0.3\ cm,\ t = 0.005\ cm$ and $\gamma = 0.198\ kgcm^{-3}$.

b) Write an R program to print the first $n = 100$ elements of the following recursive function:

$$f(n) = \frac{f(n-1)}{f(n-2)} + f(n-3),$$

where $n \geq 1$ and the initial conditions $f(1) = 0, f(2) = 1$ and $f(3) = 1$.

[End of Paper]